Indefinite Integration: Questions and Full Solutions

Question 01

English

Let $f : (0, \infty) \to (0, \infty)$ be a differentiable function with $f(1) = 1$. For every $x > 0$, the tangent to the curve $y = f(x)$ at the point $P(x, f(x))$ meets the y-axis at a point whose ordinate is $f(x)\left(1 - x^2 \log_e x\right)$. Then $f\left(\sqrt{e}\right)$ is equal to

(A) $e^{e/4}$
(B) $e^{(2e-1)/4}$
(C) $e^{1/4}$
(D) $e^{-1/4}$

Only one option is correct.

हिन्दी

माना कि $f : (0, \infty) \to (0, \infty)$ एक अवकलनीय फलन है जिसके लिए $f(1) = 1$ है। प्रत्येक $x > 0$ के लिए, वक्र $y = f(x)$ के बिन्दु $P(x, f(x))$ पर खींची गई स्पर्श रेखा y-अक्ष को ऐसे बिन्दु पर मिलती है जिसकी कोटि $f(x)\left(1 - x^2 \log_e x\right)$ है। तब $f\left(\sqrt{e}\right)$ बराबर है

(A) $e^{e/4}$
(B) $e^{(2e-1)/4}$
(C) $e^{1/4}$
(D) $e^{-1/4}$

केवल एक विकल्प सही है।

Solution

Write $y = f(x)$. The tangent at $P(x, y)$ is $Y - y = y'(X - x)$. Putting $X = 0$ gives the ordinate of its point on the y-axis: $Y = y - xy'$.

The given condition is $y - xy' = y\left(1 - x^2 \log_e x\right) = y - x^2 y \log_e x$. Cancelling y and dividing by $x > 0$ gives $y' = xy \log_e x$.

Since $y > 0$, the variables separate: $\dfrac{dy}{y} = x \log_e x \, dx$.

Integrate the right side by parts with $u = \log_e x$, $dv = x\,dx$: $\displaystyle\int x \log_e x \, dx = \frac{x^2}{2} \log_e x - \int \frac{x^2}{2} \cdot \frac{1}{x} \, dx = \frac{x^2}{2} \log_e x - \frac{x^2}{4}$.

Hence $\log_e y = \dfrac{x^2}{2} \log_e x - \dfrac{x^2}{4} + C$.

At $x = 1$: $\log_e 1 = 0 - \dfrac{1}{4} + C$, so $C = \dfrac{1}{4}$ and $f(x) = x^{x^2/2} e^{(1-x^2)/4}$.

At $x = \sqrt{e}$: $x^2 = e$ and $\log_e x = \dfrac{1}{2}$, so $\log_e f(\sqrt{e}) = \dfrac{e}{4} - \dfrac{e}{4} + \dfrac{1}{4} = \dfrac{1}{4}$.

Therefore $f(\sqrt{e}) = e^{1/4}$. Dropping the $-\dfrac{x^2}{4}$ term gives (A); a sign slip in the by-parts step gives (B).

Correct Option: (C)

Question 02

English

Let a be a real number and let $F : \mathbb{R} \to \mathbb{R}$ be a differentiable function such that $F(0) = 0$ and $F'(x) = \cos^4 x - a\cos^2 x$ for all $x \in \mathbb{R}$. If F is a periodic function, then the maximum value of F is

(A) $\dfrac{3\sqrt{3}}{64}$

(B) $\dfrac{3\sqrt{3}}{32}$

(C) $\dfrac{\sqrt{3}}{16}$

(D) $\dfrac{3}{32}$

Only one option is correct.

हिन्दी

माना कि a एक वास्तविक संख्या है तथा $F : \mathbb{R} \to \mathbb{R}$ एक अवकलनीय फलन है जिसके लिए $F(0) = 0$ तथा सभी $x \in \mathbb{R}$ के लिए $F'(x) = \cos^4 x - a\cos^2 x$ है। यदि F एक आवर्ती फलन है, तो F का अधिकतम मान है

(A) $\dfrac{3\sqrt{3}}{64}$

(B) $\dfrac{3\sqrt{3}}{32}$

(C) $\dfrac{\sqrt{3}}{16}$

(D) $\dfrac{3}{32}$

केवल एक विकल्प सही है।

Solution

Reduce the powers: $\cos^2 x = \dfrac{1+\cos 2x}{2}$ and $\cos^4 x = \left(\dfrac{1+\cos 2x}{2}\right)^2 = \dfrac{1}{4} + \dfrac{\cos 2x}{2} + \dfrac{1+\cos 4x}{8} = \dfrac{3}{8} + \dfrac{\cos 2x}{2} + \dfrac{\cos 4x}{8}$.

So $F'(x) = \left(\dfrac{3}{8} - \dfrac{a}{2}\right) + \dfrac{1-a}{2}\cos 2x + \dfrac{1}{8}\cos 4x$.

Integrating with $F(0) = 0$: $F(x) = \left(\dfrac{3}{8} - \dfrac{a}{2}\right)x + \dfrac{1-a}{4}\sin 2x + \dfrac{1}{32}\sin 4x$.

The trigonometric part is bounded and periodic, while a non-zero multiple of x is unbounded. So F is periodic exactly when $\dfrac{3}{8} - \dfrac{a}{2} = 0$, i.e. $a = \dfrac{3}{4}$.

With $a = \dfrac{3}{4}$: $F(x) = \dfrac{1}{16}\sin 2x + \dfrac{1}{32}\sin 4x = \dfrac{\sin 2x}{16}(1+\cos 2x) = \dfrac{2\sin x\cos x \cdot 2\cos^2 x}{16} = \dfrac{1}{4}\sin x\cos^3 x$.

$F'(x) = \dfrac{1}{4}\left(\cos^4 x - 3\sin^2 x\cos^2 x\right) = \dfrac{1}{4}\cos^2 x\left(\cos^2 x - 3\sin^2 x\right)$, which vanishes when $\cos x = 0$ or $\tan^2 x = \dfrac{1}{3}$.

F has period π. At $\cos x = 0$, $F = 0$. At $\tan x = \dfrac{1}{\sqrt{3}}$ $\left(x = \dfrac{\pi}{6}\right)$, $F = \dfrac{1}{4}\cdot\dfrac{1}{2}\cdot\left(\dfrac{\sqrt{3}}{2}\right)^3 = \dfrac{3\sqrt{3}}{64} > 0$; at $x = -\dfrac{\pi}{6}$ the value is $-\dfrac{3\sqrt{3}}{64}$.

Hence the maximum value of F is $\dfrac{3\sqrt{3}}{64}$. Option (B) forgets the factor $\dfrac{1}{2}$ from $\sin x$; option (C) is the value of the first term alone at the wrong point.

Correct Option: (A)

Question 03

English

Let p and q be real numbers and let $F : \mathbb{R} \to \mathbb{R}$ be a differentiable function such that $F(-2) = 0$ and $F'(x) = \dfrac{px + q}{x^2 + 4x + 13}$ for all $x \in \mathbb{R}$. Suppose that the function $F(x) - \log_e\left(x^2 + 4x + 13\right)$ is bounded on $\mathbb{R}$ and that $\lim\limits_{x\to\infty}(F(x) - F(-x)) = \pi$. Then $F(1)$ is equal to

(A) $\log_e 2 + \dfrac{\pi}{2}$

(B) $\log_e 2 + \dfrac{\pi}{4}$

(C) $2\log_e 2 + \dfrac{\pi}{4}$

(D) $\log_e 2 + \dfrac{\pi}{12}$

Only one option is correct.

हिन्दी

माना कि p तथा q वास्तविक संख्याएँ हैं तथा $F : \mathbb{R} \to \mathbb{R}$ एक अवकलनीय फलन है जिसके लिए $F(-2) = 0$ तथा सभी $x \in \mathbb{R}$ के लिए $F'(x) = \dfrac{px + q}{x^2 + 4x + 13}$ है। माना कि फलन $F(x) - \log_e\left(x^2 + 4x + 13\right)$, $\mathbb{R}$ पर परिबद्ध है तथा $\lim\limits_{x\to\infty}(F(x) - F(-x)) = \pi$ है। तब $F(1)$ बराबर है

(A) $\log_e 2 + \dfrac{\pi}{2}$

(B) $\log_e 2 + \dfrac{\pi}{4}$

(C) $2\log_e 2 + \dfrac{\pi}{4}$

(D) $\log_e 2 + \dfrac{\pi}{12}$

केवल एक विकल्प सही है।

Solution

Write the numerator in terms of the derivative of the denominator: $px + q = \dfrac{p}{2}(2x + 4) + (q - 2p)$.

Then $F'(x) = \dfrac{p}{2}\cdot\dfrac{2x + 4}{x^2 + 4x + 13} + \dfrac{q - 2p}{(x + 2)^2 + 3^2}$, so $F(x) = \dfrac{p}{2}\log_e\left(x^2 + 4x + 13\right) + \dfrac{q - 2p}{3}\tan^{-1}\dfrac{x + 2}{3} + C$.

$F(x) - \log_e\left(x^2 + 4x + 13\right) = \left(\dfrac{p}{2} - 1\right)\log_e\left(x^2 + 4x + 13\right) + \dfrac{q - 2p}{3}\tan^{-1}\dfrac{x + 2}{3} + C$. The $\tan^{-1}$ term is bounded and the log term is unbounded as $x \to \infty$, so boundedness forces $\dfrac{p}{2} = 1$, i.e. $p = 2$.

Now $F(x) - F(-x) = \log_e\dfrac{x^2 + 4x + 13}{x^2 - 4x + 13} + \dfrac{q - 4}{3}\left(\tan^{-1}\dfrac{x + 2}{3} - \tan^{-1}\dfrac{2 - x}{3}\right)$.

As $x \to \infty$ the ratio inside the logarithm tends to 1, while $\tan^{-1}\dfrac{x + 2}{3} \to \dfrac{\pi}{2}$ and $\tan^{-1}\dfrac{2 - x}{3} \to -\dfrac{\pi}{2}$.

So the limit is $\dfrac{q - 4}{3}\pi$.

Setting it equal to π: $q - 4 = 3$, so $q = 7$, and $F(x) = \log_e\left(x^2 + 4x + 13\right) + \tan^{-1}\dfrac{x + 2}{3} + C$.

$F(-2) = \log_e 9 + 0 + C = 0$ gives $C = -\log_e 9$.
$F(1) = \log_e 18 + \tan^{-1} 1 - \log_e 9 = \log_e 2 + \frac{\pi}{4}$.

Correct Option: (B)

Question 04

English

Let $F : \left(0, \frac{\pi}{2}\right) \to \mathbb{R}$ be a differentiable function such that $F'(x) = \frac{7\cos x - \sin x}{3\sin x + 4\cos x}$ for all $x \in \left(0, \frac{\pi}{2}\right)$ and $\lim\limits_{x \to 0^+} F(x) = \log_e 4$. Then the maximum value of F on $\left(0, \frac{\pi}{2}\right)$ is

(A) $\tan^{-1} 7 + \log_e 5$
(B) $\tan^{-1} \frac{1}{7} + \log_e \frac{5}{\sqrt{2}}$
(C) $\tan^{-1} 7 - \log_e \frac{5}{\sqrt{2}}$
(D) $\tan^{-1} 7 + \log_e \frac{5}{\sqrt{2}}$

Only one option is correct.

हिन्दी

माना कि $F : \left(0, \frac{\pi}{2}\right) \to \mathbb{R}$ एक अवकलनीय फलन है जिसके लिए सभी $x \in \left(0, \frac{\pi}{2}\right)$ के लिए $F'(x) = \frac{7\cos x - \sin x}{3\sin x + 4\cos x}$ तथा $\lim\limits_{x \to 0^+} F(x) = \log_e 4$ है। तब $\left(0, \frac{\pi}{2}\right)$ पर F का अधिकतम मान है

(A) $\tan^{-1} 7 + \log_e 5$
(B) $\tan^{-1} \frac{1}{7} + \log_e \frac{5}{\sqrt{2}}$
(C) $\tan^{-1} 7 - \log_e \frac{5}{\sqrt{2}}$
(D) $\tan^{-1} 7 + \log_e \frac{5}{\sqrt{2}}$

केवल एक विकल्प सही है।

Solution

On $\left(0, \frac{\pi}{2}\right)$ the denominator $D = 3\sin x + 4\cos x$ is positive, and $D' = 3\cos x - 4\sin x$.
Write the numerator as $A\,D + B\,D'$: comparing coefficients of $\sin x$ and $\cos x$ gives $3A - 4B = -1$ and $4A + 3B = 7$, so $A = 1$, $B = 1$.
Hence $F'(x) = 1 + \frac{D'}{D}$ and $F(x) = x + \log_e(3\sin x + 4\cos x) + C$.
As $x \to 0^+$, $F(x) \to \log_e 4 + C$, so $C = 0$.
$F'(x) = 0$ exactly when $7\cos x = \sin x$, i.e. $\tan x = 7$. For $\tan x < 7$ the numerator $7\cos x - \sin x$ is positive and for $\tan x > 7$ it is negative, so F increases and then decreases: its maximum is at $x_0 = \tan^{-1} 7$.
With $\tan x_0 = 7$: $\sin x_0 = \frac{7}{\sqrt{50}}$, $\cos x_0 = \frac{1}{\sqrt{50}}$, so $3\sin x_0 + 4\cos x_0 = \frac{25}{5\sqrt{2}} = \frac{5}{\sqrt{2}}$.
Therefore the maximum value is $\tan^{-1} 7 + \log_e \frac{5}{\sqrt{2}}$. Option (B) uses the wrong critical point and (C) the wrong sign of B.

Correct Option: (D)

Question 05

English

Let $F : (0, \infty) \to \mathbb{R}$ be a differentiable function such that $F'(x) = \dfrac{1}{x + \sqrt{x^2 + 2x}}$ for all $x > 0$ and $\lim\limits_{x \to 0^+} F(x) = 0$. Then $\lim\limits_{x \to \infty} \left(F(x) - \dfrac{1}{2} \log_e x \right)$ is equal to

(A) $\dfrac{1 + \log_e 2}{2}$

(B) $\dfrac{\log_e 2}{2}$

(C) $\dfrac{1 - \log_e 2}{2}$

(D) $1 + \log_e 2$

Only one option is correct.

हिन्दी

माना कि $F : (0, \infty) \to \mathbb{R}$ एक अवकलनीय फलन है जिसके लिए सभी $x > 0$ के लिए $F'(x) = \dfrac{1}{x + \sqrt{x^2 + 2x}}$ तथा $\lim\limits_{x \to 0^+} F(x) = 0$ है। तब $\lim\limits_{x \to \infty} \left(F(x) - \dfrac{1}{2} \log_e x \right)$ बराबर है

(A) $\dfrac{1 + \log_e 2}{2}$

(B) $\dfrac{\log_e 2}{2}$

(C) $\dfrac{1 - \log_e 2}{2}$

(D) $1 + \log_e 2$

केवल एक विकल्प सही है।

Solution

Use the Euler substitution $\sqrt{x^2 + 2x} = t - x$ (with $t = x + \sqrt{x^2 + 2x} > 0$). Squaring: $x^2 + 2x = t^2 - 2tx + x^2$, so $x = \dfrac{t^2}{2(t+1)}$.

Then $dx = \dfrac{2t \cdot 2(t+1) - 2t^2}{4(t+1)^2}\, dt = \dfrac{t(t+2)}{2(t+1)^2}\, dt$, and the integrand is $\dfrac{1}{t}$.

So $\displaystyle\int \frac{dx}{x + \sqrt{x^2 + 2x}} = \int \frac{t+2}{2(t+1)^2}\, dt = \frac{1}{2} \int \left(\frac{1}{t+1} + \frac{1}{(t+1)^2} \right) dt = \frac{1}{2} \log_e (t+1) - \frac{1}{2(t+1)} + C.$

Back in x: $t + 1 = x + 1 + \sqrt{x^2 + 2x}$ and $\dfrac{1}{t+1} = \dfrac{x + 1 - \sqrt{x^2 + 2x}}{(x+1)^2 - (x^2 + 2x)} = x + 1 - \sqrt{x^2 + 2x}$.

Hence $F(x) = \dfrac{1}{2} \log_e \left(x + 1 + \sqrt{x^2 + 2x} \right) + \dfrac{\sqrt{x^2 + 2x} - x - 1}{2} + C'$; absorbing $-\dfrac{1}{2}$ into the constant,

$F(x) = \dfrac{1}{2} \log_e \left(x + 1 + \sqrt{x^2 + 2x} \right) + \dfrac{\sqrt{x^2 + 2x} - x}{2} + C.$

As $x \to 0^+$ both terms tend to 0, so $C = 0$.

As $x \to \infty$: $\dfrac{1}{2} \log_e \left(x + 1 + \sqrt{x^2 + 2x} \right) - \dfrac{1}{2} \log_e x = \dfrac{1}{2} \log_e \left(1 + \dfrac{1}{x} + \sqrt{1 + \dfrac{2}{x}} \right) \to \dfrac{1}{2} \log_e 2.$

Also $\sqrt{x^2+2x}-x=\dfrac{2x}{\sqrt{x^2+2x}+x}\to 1$, so the second term tends to $\dfrac{1}{2}$.

The limit is $\dfrac{1+\log_e 2}{2}$. Option (B) drops the algebraic term and (C) has its sign reversed.

Correct Option: (A)

Question 06

English

Let $F:(0,\infty)\to\mathbb{R}$ be a differentiable function such that $F'(x)=\dfrac{1}{x^2\sqrt{x^2+x+1}}$ for all $x>0$ and $\lim\limits_{x\to\infty}F(x)=0$. Then $F(1)$ is equal to

(A) $1-\sqrt{3}-\dfrac{1}{2}\log_e\left(1+\dfrac{2}{\sqrt{3}}\right)$

(B) $1-\sqrt{3}+\dfrac{1}{2}\log_e\left(1+\dfrac{2}{\sqrt{3}}\right)$

(C) $\sqrt{3}-1+\dfrac{1}{2}\log_e\left(1+\dfrac{2}{\sqrt{3}}\right)$

(D) $1-\sqrt{3}+\log_e\left(1+\dfrac{2}{\sqrt{3}}\right)$

Only one option is correct.

हिन्दी

माना कि $F:(0,\infty)\to\mathbb{R}$ एक अवकलनीय फलन है जिसके लिए सभी $x>0$ के लिए $F'(x)=\dfrac{1}{x^2\sqrt{x^2+x+1}}$ तथा $\lim\limits_{x\to\infty}F(x)=0$ है। तब $F(1)$ बराबर है

(A) $1-\sqrt{3}-\dfrac{1}{2}\log_e\left(1+\dfrac{2}{\sqrt{3}}\right)$

(B) $1-\sqrt{3}+\dfrac{1}{2}\log_e\left(1+\dfrac{2}{\sqrt{3}}\right)$

(C) $\sqrt{3}-1+\dfrac{1}{2}\log_e\left(1+\dfrac{2}{\sqrt{3}}\right)$

(D) $1-\sqrt{3}+\log_e\left(1+\dfrac{2}{\sqrt{3}}\right)$

केवल एक विकल्प सही है।

Solution

For $x>0$ put $x=\dfrac{1}{t}$ with $t>0$. Then $x^2+x+1=\dfrac{1+t+t^2}{t^2}$, so $\sqrt{x^2+x+1}=\dfrac{\sqrt{t^2+t+1}}{t}$, and $dx=-\dfrac{dt}{t^2}$.

Hence $\dfrac{dx}{x^2\sqrt{x^2+x+1}}=t^2\cdot\dfrac{t}{\sqrt{t^2+t+1}}\cdot\left(-\dfrac{dt}{t^2}\right)=-\dfrac{t\,dt}{\sqrt{t^2+t+1}}$.

Since $\dfrac{d}{dt}\sqrt{t^2+t+1}=\dfrac{2t+1}{2\sqrt{t^2+t+1}}$, write $t=\dfrac{1}{2}(2t+1)-\dfrac{1}{2}$.

Then $\displaystyle\int\frac{t\,dt}{\sqrt{t^2+t+1}}=\sqrt{t^2+t+1}-\frac{1}{2}\int\frac{dt}{\sqrt{\left(t+\frac{1}{2}\right)^2+\frac{3}{4}}}=\sqrt{t^2+t+1}-\frac{1}{2}\log_e\left(t+\frac{1}{2}+\sqrt{t^2+t+1}\right)$.

So $F = -\sqrt{t^2+t+1} + \frac{1}{2}\log_e\left(t + \frac{1}{2} + \sqrt{t^2+t+1}\right) + C$ with $t = \frac{1}{x}$.

As $x \to \infty$, $t \to 0^+$ and $F \to -1 + \frac{1}{2}\log_e \frac{3}{2} + C$. This must be 0, so $C = 1 - \frac{1}{2}\log_e \frac{3}{2}$.

At $x = 1$ ($t = 1$): $F(1) = -\sqrt{3} + \frac{1}{2}\log_e\left(\frac{3}{2} + \sqrt{3}\right) + 1 - \frac{1}{2}\log_e \frac{3}{2} = 1 - \sqrt{3} + \frac{1}{2}\log_e\left(1 + \frac{2}{\sqrt{3}}\right)$.

Check: $F' > 0$ and $F \to 0$ at infinity, so $F(1) < 0$; indeed $1 - \sqrt{3} + \frac{1}{2}\log_e(2.155) \approx -0.348$. Options (A) and (C) carry sign slips and (D) loses the factor $\frac{1}{2}$.

Correct Option: (B)

Question 07

English

Let $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ be a differentiable function such that $f'(x) = \frac{3x^4 - 2}{x^2}$ for all $x \neq 0$. Suppose that $f(1) = 7$ and $f(x) + f(-x) = 4$ for all $x \neq 0$. Then the value of $f(-2)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ एक अवकलनीय फलन है जिसके लिए सभी $x \neq 0$ के लिए $f'(x) = \frac{3x^4 - 2}{x^2}$ है। माना कि $f(1) = 7$ तथा सभी $x \neq 0$ के लिए $f(x) + f(-x) = 4$ है। तब $f(-2)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Split the integrand: $f'(x) = \frac{3x^4 - 2}{x^2} = 3x^2 - 2x^{-2}$. By the power rule an antiderivative is $x^3 + \frac{2}{x}$. The domain $\mathbb{R} \setminus \{0\}$ is the union of the two intervals $(-\infty, 0)$ and $(0, \infty)$. On each interval two antiderivatives differ by a constant, but the constants on the two intervals are independent. So the general solution of $y' = 3x^2 - \frac{2}{x^2}$ on this domain is

$f(x) = x^3 + \frac{2}{x} + C_1$ for $x > 0$ and $f(x) = x^3 + \frac{2}{x} + C_2$ for $x < 0$.

The condition $f(1) = 7$ gives $1 + 2 + C_1 = 7$, so $C_1 = 4$.

For $x > 0$: $f(-x) = -x^3 - \frac{2}{x} + C_2$, so $f(x) + f(-x) = C_1 + C_2$ for every x. The condition gives $C_1 + C_2 = 4$, so $C_2 = 0$.

Verification: $f(x) = x^3 + \frac{2}{x} + 4$ ($x > 0$), $f(x) = x^3 + \frac{2}{x}$ ($x < 0$) satisfies the differential equation on both intervals, $f(1) = 7$, and $f(x) + f(-x) = 4$.

Hence $f(-2) = -8 + \frac{2}{-2} + 0 = -9$.

Using one constant for the whole domain would force $2C = 4$ and $C = 4$ at the same time, a contradiction, which shows why two constants are needed.

Final Answer: -9

Question 08

English

Let $f : [0, \infty) \to [0, \infty)$ be a continuous function with $f(0) = 0$ and $f(x) > 0$ for $x > 0$, which is differentiable on $(0, \infty)$. For each $x > 0$, let M be the foot of the perpendicular from the point $P(x, f(x))$ to the x-axis, and let the normal to the curve $y = f(x)$ at P meet the x-axis at N. Suppose that $MN = 3x\sqrt{x^2 + 9}$ for every $x > 0$. Then the value of $f(4)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : [0, \infty) \to [0, \infty)$ एक सतत फलन है जिसके लिए $f(0) = 0$ तथा $x > 0$ के लिए $f(x) > 0$ है, और f, $(0, \infty)$ पर अवकलनीय है। प्रत्येक $x > 0$ के लिए, माना कि बिन्दु $P(x, f(x))$ से x-अक्ष पर डाले गए लम्ब का पाद M है, तथा P पर वक्र $y = f(x)$ का अभिलम्ब x-अक्ष को N पर मिलता है। माना कि प्रत्येक $x > 0$ के लिए $MN = 3x\sqrt{x^2 + 9}$ है। तब $f(4)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Write $y = f(x)$. The normal at $P(x, y)$ is $X - x + y'(Y - y) = 0$. Putting $Y = 0$ gives $X = x + yy'$, so $N = (x + yy', 0)$ while $M = (x, 0)$. Hence $MN = |yy'|$.

The condition is $|yy'| = 3x\sqrt{x^2 + 9}$, which is positive for every $x > 0$. Since $yy' = \dfrac{1}{2}\dfrac{d}{dx}\left(y^2\right)$ is continuous on $(0, \infty)$ and never zero, it has one sign there.

If it were negative, y^2 would be strictly decreasing on $(0, \infty)$; but $y^2 \to 0$ as $x \to 0^+$ and $y^2 > 0$ for $x > 0$, which is impossible. So $yy' = 3x\sqrt{x^2 + 9}$.

Separate the variables: $y\,dy = 3x\sqrt{x^2 + 9}\,dx$.

Substitute $u = x^2 + 9$, $du = 2x\,dx$: $\displaystyle\int 3x\sqrt{x^2 + 9}\,dx = \frac{3}{2}\int u^{1/2}\,du = u^{3/2} = \left(x^2 + 9\right)^{3/2}$.

So $\dfrac{y^2}{2} = \left(x^2 + 9\right)^{3/2} + C$. Letting $x \to 0^+$ and using $f(0) = 0$: $0 = 27 + C$, so $C = -27$.

Thus $f(x)^2 = 2\left(x^2 + 9\right)^{3/2} - 54$ and $f(x) > 0$ for $x > 0$.

At $x = 4$: $x^2 + 9 = 25$, $(25)^{3/2} = 125$, so $f(4)^2 = 250 - 54 = 196$ and $f(4) = 14$.

Final Answer: 14

Question 09

English

Let $F : (-3, 3) \to \mathbb{R}$ be a differentiable function such that $F(0) = 0$ and $F'(x) = \dfrac{x^2}{\sqrt{9 - x^2}}$ for all $x \in (-3, 3)$. Let θ be the angle with $-90^\circ < \theta < 90^\circ$ such that $F(3 \sin \theta) = \dfrac{9\sqrt{2}}{8} - \dfrac{9\pi}{16}$. If θ is measured in degrees, then the value of θ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $F : (-3, 3) \to \mathbb{R}$ एक अवकलनीय फलन है जिसके लिए $F(0) = 0$ तथा सभी $x \in (-3, 3)$ के लिए $F'(x) = \dfrac{x^2}{\sqrt{9 - x^2}}$ है। माना कि θ वह कोण है जिसके लिए $-90^\circ < \theta < 90^\circ$ तथा $F(3 \sin \theta) = \dfrac{9\sqrt{2}}{8} - \dfrac{9\pi}{16}$ है। यदि θ को अंशों (degrees) में मापा जाए, तो θ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Use the substitution $x = 3 \sin \phi$ with $\phi \in \left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$ measured in radians (calculus formulas require radians). Then $\cos \phi > 0$, $\sqrt{9 - x^2} = 3 \cos \phi$ and $dx = 3 \cos \phi \, d\phi$.

The integrand becomes $\dfrac{9 \sin^2 \phi}{3 \cos \phi} \cdot 3 \cos \phi \, d\phi = 9 \sin^2 \phi \, d\phi = \dfrac{9}{2} (1 - \cos 2\phi) \, d\phi$.

Integrating: $\dfrac{9}{2}\phi - \dfrac{9}{4} \sin 2\phi + C$. Since $F(0) = 0$ corresponds to $\phi = 0$, $C = 0$ and $F(3 \sin \phi) = \dfrac{9}{2}\phi - \dfrac{9}{4} \sin 2\phi$. (Geometrically, $\dfrac{9}{2} \cdot 2\phi$ is the area of a sector of radius 3 and angle 2ϕ, so F is half a sector minus a triangle.)

Uniqueness: $F'(x) \geq 0$ with equality only at $x = 0$, so F is strictly increasing; $\phi \mapsto 3 \sin \phi$ is strictly increasing, so the composition is strictly increasing and the equation has at most one root.

Compare $\dfrac{9}{2}\phi - \dfrac{9}{4} \sin 2\phi$ with $-\dfrac{9\pi}{16} + \dfrac{9\sqrt{2}}{8}$: the choice $\phi = -\dfrac{\pi}{8}$ gives $\dfrac{9}{2}\phi = -\dfrac{9\pi}{16}$ and $\sin 2\phi = \sin\left(-\dfrac{\pi}{4}\right) = -\dfrac{\sqrt{2}}{2}$, so $-\dfrac{9}{4} \sin 2\phi = \dfrac{9\sqrt{2}}{8}$. It is a root, hence the only one.

Convert to degrees with π rad $= 180^\circ$: $\theta = -\dfrac{\pi}{8} \cdot \dfrac{180^\circ}{\pi} = -22.5^\circ$.

Final Answer: -22.50

Question 10

English

Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function such that $y = f(x)$ satisfies the differential equation $\left(1+x^2\right)^2\left(\frac{dy}{dx}+y\right) = x^3 - x^2 + x + 1$ for all $x \in \mathbb{R}$, and f is bounded on $(-\infty, 0]$. Then the maximum value of f on $\mathbb{R}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$ एक अवकलनीय फलन है जिसके लिए $y = f(x)$ सभी $x \in \mathbb{R}$ के लिए अवकल समीकरण $\left(1+x^2\right)^2\left(\frac{dy}{dx}+y\right) = x^3 - x^2 + x + 1$ को सन्तुष्ट करता है, तथा f, $(-\infty, 0]$ पर परिबद्ध है। तब $\mathbb{R}$ पर f का अधिकतम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Divide by $(1+x^2)^2 > 0$: $\frac{dy}{dx} + y = Q(x)$ with $Q(x) = \frac{x^3 - x^2 + x + 1}{(1+x^2)^2}$. This is linear with $P = 1$, so the integrating factor is $e^{\int 1\,dx} = e^x$ and $\frac{d}{dx}\left(ye^x\right) = e^x Q(x)$.

Split the numerator: $x^3 - x^2 + x + 1 = x(1+x^2) + (1-x^2)$. So $Q(x) = \frac{x}{1+x^2} + \frac{1-x^2}{(1+x^2)^2}$.

With $g(x) = \frac{x}{1+x^2}$ we have $g'(x) = \frac{(1+x^2) - 2x^2}{(1+x^2)^2} = \frac{1-x^2}{(1+x^2)^2}$, so $Q = g + g'$.

Since $\frac{d}{dx}\left(e^x g(x)\right) = e^x\left(g + g'\right)$, we get $\int e^x Q(x)\,dx = e^x g(x) + C$.

Hence $ye^x = \frac{xe^x}{1+x^2} + C$, i.e. $f(x) = \frac{x}{1+x^2} + Ce^{-x}$ (the general solution).

As $x \to -\infty$, $\frac{x}{1+x^2} \to 0$ while $e^{-x} \to \infty$. So f is bounded on $(-\infty, 0]$ only if $C = 0$.

Thus $f(x) = \frac{x}{1+x^2}$. For $x \le 0$, $f \le 0$; for $x > 0$, $1 + x^2 \ge 2x$ gives $f(x) \le \frac{1}{2}$, with equality at $x = 1$.

The maximum value of f is $\frac{1}{2} = 0.50$.

Final Answer: 0.50

Question 11

English

Let $F : \mathbb{R} \to \mathbb{R}$ and $G : \mathbb{R} \to \mathbb{R}$ be differentiable functions such that $F'(x) = e^{5x} \sin 12x$ and $G'(x) = F(x)$ for all $x \in \mathbb{R}$, and $\lim\limits_{x \to -\infty} F(x) = \lim\limits_{x \to -\infty} G(x) = 0$. Let M be the maximum value of $e^{-5x}G(x)$ on $\mathbb{R}$. Then the value of $\dfrac{1}{M}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $F : \mathbb{R} \to \mathbb{R}$ तथा $G : \mathbb{R} \to \mathbb{R}$ अवकलनीय फलन हैं जिनके लिए सभी $x \in \mathbb{R}$ के लिए $F'(x) = e^{5x} \sin 12x$ तथा $G'(x) = F(x)$ हैं, और $\lim\limits_{x \to -\infty} F(x) = \lim\limits_{x \to -\infty} G(x) = 0$ है। माना कि $\mathbb{R}$ पर $e^{-5x}G(x)$ का अधिकतम मान M है। तब $\dfrac{1}{M}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $I = \displaystyle\int e^{5x} \sin 12x \, dx$ and $J = \displaystyle\int e^{5x} \cos 12x \, dx$. Integrating by parts with e^{5x} as the function to integrate:

$I = \dfrac{e^{5x} \sin 12x}{5} - \dfrac{12}{5} J$ and $J = \dfrac{e^{5x} \cos 12x}{5} + \dfrac{12}{5} I$.

Substituting J into I: $I = \dfrac{e^{5x} \sin 12x}{5} - \dfrac{12e^{5x} \cos 12x}{25} - \dfrac{144}{25} I$, so $\dfrac{169}{25} I = \dfrac{e^{5x}(5 \sin 12x - 12 \cos 12x)}{25}$ and $I = \dfrac{e^{5x}(5 \sin 12x - 12 \cos 12x)}{169}$. Similarly $J = \dfrac{e^{5x}(5 \cos 12x + 12 \sin 12x)}{169}$.

So $F(x) = \dfrac{e^{5x}(5 \sin 12x - 12 \cos 12x)}{169} + C_1$. The bracket is bounded and $e^{5x} \to 0$ as $x \to -\infty$, so $\lim\limits_{x \to -\infty} F = C_1 = 0$.

Then $G(x) = \dfrac{1}{169} (5I - 12J) + C_2 = \dfrac{e^{5x}}{169^2} (25 \sin 12x - 60 \cos 12x - 60 \cos 12x - 144 \sin 12x) + C_2$.

Hence $G(x) = \dfrac{e^{5x}(-119 \sin 12x - 120 \cos 12x)}{28561} + C_2$, and the same limit argument gives $C_2 = 0$.

$e^{-5x}G(x) = \dfrac{-119 \sin 12x - 120 \cos 12x}{28561}$. The maximum of $a \sin \theta + b \cos \theta$ is $\sqrt{a^2 + b^2}$, and $\sqrt{119^2 + 120^2} = \sqrt{28561} = 169$.

So $M = \dfrac{169}{28561} = \dfrac{1}{169}$ and $\dfrac{1}{M} = 169$.

Final Answer: 169

Question 12

English

Let $F : (0, \infty) \to \mathbb{R}$ be a differentiable function such that $F'(x) = -\dfrac{3x+2}{x(x+1)(x+2)}$ for all $x > 0$ and $\lim\limits_{x\to\infty} F(x) = 0$. Then the value of $e^{F(1)}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $F : (0, \infty) \to \mathbb{R}$ एक अवकलनीय फलन है जिसके लिए सभी $x > 0$ के लिए $F'(x) = -\dfrac{3x+2}{x(x+1)(x+2)}$ तथा $\lim\limits_{x\to\infty} F(x) = 0$ है। तब $e^{F(1)}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Decompose $\dfrac{3x+2}{x(x+1)(x+2)} = \dfrac{A}{x} + \dfrac{B}{x+1} + \dfrac{C}{x+2}$, so $3x+2 = A(x+1)(x+2)+Bx(x+2)+Cx(x+1)$.
Put $x = 0$: $2 = 2A$, $A = 1$. Put $x = -1$: $-1 = -B$, $B = 1$. Put $x = -2$: $-4 = 2C$, $C = -2$.
So $F'(x) = -\dfrac{1}{x} - \dfrac{1}{x+1} + \dfrac{2}{x+2}$ and, for $x > 0$, $F(x) = -\log_e x - \log_e(x+1) + 2\log_e(x+2) + K = \log_e \dfrac{(x+2)^2}{x(x+1)} + K$.

The sum of the coefficients $-1 - 1 + 2$ is 0, which is why F has a finite limit: $\dfrac{(x+2)^2}{x(x+1)} = \dfrac{x^2+4x+4}{x^2+x} \to 1$, so $F(x) \to K$ and $K = 0$.
Then $F(1) = \log_e \dfrac{9}{2}$ and $e^{F(1)} = \dfrac{9}{2} = 4.5$.

Final Answer: 4.50

Question 13

English

Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function such that $f(-2) = 1$ and $y = f(x)$ satisfies the differential equation $2\left(x^2 + 4x + 13\right)\dfrac{dy}{dx} = 3\left(y^2 - 2y + 2\right)$ for all $x \in \mathbb{R}$. Then the value of $f(2)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$ एक अवकलनीय फलन है जिसके लिए $f(-2) = 1$ है तथा $y = f(x)$ सभी $x \in \mathbb{R}$ के लिए अवकल समीकरण $2\left(x^2 + 4x + 13\right)\dfrac{dy}{dx} = 3\left(y^2 - 2y + 2\right)$ को सन्तुष्ट करता है। तब $f(2)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Complete the squares: $y^2 - 2y + 2 = (y-1)^2 + 1 > 0$ and $x^2 + 4x + 13 = (x+2)^2 + 9 > 0$.

The variables separate: $\dfrac{dy}{(y-1)^2 + 1} = \dfrac{3}{2} \cdot \dfrac{dx}{(x+2)^2 + 3^2}$.

Using $\displaystyle\int \frac{du}{u^2 + a^2} = \frac{1}{a}\tan^{-1}\frac{u}{a}$: $\tan^{-1}(y-1) = \dfrac{3}{2} \cdot \dfrac{1}{3}\tan^{-1}\dfrac{x+2}{3} + C = \dfrac{1}{2}\tan^{-1}\dfrac{x+2}{3} + C$.

At $x = -2$, $y = 1$: $0 = 0 + C$, so $C = 0$. The right side lies in $\left(-\dfrac{\pi}{4}, \dfrac{\pi}{4}\right)$, so $y = 1 + \tan\left(\dfrac{1}{2}\tan^{-1}\dfrac{x+2}{3}\right)$ is defined for all x.

At $x = 2$: let $\phi = \tan^{-1}\dfrac{4}{3} \in \left(0, \dfrac{\pi}{2}\right)$; then $\sin\phi = \dfrac{4}{5}$, $\cos\phi = \dfrac{3}{5}$.

$\tan\dfrac{\phi}{2} = \dfrac{\sin\phi}{1 + \cos\phi} = \dfrac{4/5}{8/5} = \dfrac{1}{2}$.

Hence $f(2) = 1 + \dfrac{1}{2} = 1.5$.

Final Answer: 1.50

Question 14

English

Let $k > -2$ be a real number and let $F : (0, \infty) \to \mathbb{R}$ be a differentiable function such that $F(1) = 0$ and $F'(x) = \dfrac{x^2 + 1}{x^4 + kx^2 + 1}$ for all $x > 0$. If $\lim\limits_{x \to \infty} F(x) = \pi$, then the value of k is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $k > -2$ एक वास्तविक संख्या है तथा $F : (0, \infty) \to \mathbb{R}$ एक अवकलनीय फलन है जिसके लिए $F(1) = 0$ तथा सभी $x > 0$ के लिए $F'(x) = \dfrac{x^2 + 1}{x^4 + kx^2 + 1}$ है। यदि $\lim\limits_{x \to \infty} F(x) = \pi$ है, तो k का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For $x > 0$ divide numerator and denominator by x^2: $F'(x) = \dfrac{1 + \frac{1}{x^2}}{x^2 + \frac{1}{x^2} + k}$.

Since $x^2 + \dfrac{1}{x^2} = \left(x - \dfrac{1}{x}\right)^2 + 2$, the denominator is $\left(x - \dfrac{1}{x}\right)^2 + k + 2$, which is positive because $k > -2$.

Substitute $t = x - \dfrac{1}{x}$, $dt = \left(1 + \dfrac{1}{x^2}\right) dx$. With $a = \sqrt{k+2} > 0$ the integral is $\displaystyle\int \frac{dt}{t^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{t}{a} + C$.

So $F(x) = \dfrac{1}{a} \tan^{-1} \dfrac{x^2 - 1}{ax} + C$. At $x = 1$, $t = 0$, so $F(1) = C = 0$.

As $x \to \infty$, $t = x - \dfrac{1}{x} \to \infty$ and $\tan^{-1} \dfrac{t}{a} \to \dfrac{\pi}{2}$, so $\lim\limits_{x \to \infty} F(x) = \dfrac{\pi}{2a}$.

$\dfrac{\pi}{2a} = \pi$ gives $a = \dfrac{1}{2}$, so $k + 2 = \dfrac{1}{4}$ and $k = -\dfrac{7}{4}$.

Check: with $k = -\dfrac{7}{4}$, $x^4 - \dfrac{7}{4}x^2 + 1 = \left(x^2 - \dfrac{7}{8}\right)^2 + \dfrac{15}{64} > 0$, so F' is defined everywhere.

Final Answer: -1.75

Question 15

English

Let a and b be real numbers with $a > 0$ and $a + b > 0$. Let $F : \left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right) \to \mathbb{R}$ and $G : (-\pi, \pi) \to \mathbb{R}$ be differentiable functions such that $F(0) = G(0) = 0$,

$$F'(x) = \frac{1}{a + b \sin^2 x} \quad \text{and} \quad G'(x) = \frac{1}{5 + 3 \cos x}$$

for all x in their respective domains. If $F\left(\dfrac{\pi}{4}\right) = G\left(\dfrac{\pi}{2}\right)$ and $F\left(\dfrac{\pi}{3}\right) = G\left(\dfrac{2\pi}{3}\right)$, then the value of $a^2 + b^2$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि a तथा b वास्तविक संख्याएँ हैं जिनके लिए $a > 0$ तथा $a + b > 0$ है। माना कि $F : \left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right) \to \mathbb{R}$ तथा

$G : (-\pi, \pi) \to \mathbb{R}$ अवकलनीय फलन हैं जिनके लिए $F(0) = G(0) = 0$ है, तथा अपने-अपने प्रांत के सभी x के लिए

$$F'(x) = \frac{1}{a + b\sin^2 x}$$

तथा

$$G'(x) = \frac{1}{5 + 3\cos x}$$

हैं। यदि $F\left(\frac{\pi}{4}\right) = G\left(\frac{\pi}{2}\right)$ तथा $F\left(\frac{\pi}{3}\right) = G\left(\frac{2\pi}{3}\right)$ है, तो $a^2 + b^2$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

For G on $(-\pi, \pi)$ put $t = \tan\frac{x}{2}$: $\cos x = \frac{1 - t^2}{1 + t^2}$ and $dx = \frac{2\,dt}{1 + t^2}$, so $\frac{dx}{5 + 3\cos x} = \frac{2\,dt}{5(1 + t^2) + 3(1 - t^2)} = \frac{2\,dt}{8 + 2t^2} = \frac{dt}{4 + t^2}$.

Hence $G(x) = \frac{1}{2}\tan^{-1}\left(\frac{1}{2}\tan\frac{x}{2}\right)$ (with $G(0) = 0$). So $G\left(\frac{\pi}{2}\right) = \frac{1}{2}\tan^{-1}\frac{1}{2}$ and $G\left(\frac{2\pi}{3}\right) = \frac{1}{2}\tan^{-1}\frac{\sqrt{3}}{2}$.

For F on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ divide numerator and denominator by $\cos^2 x$: $\frac{\sec^2 x\,dx}{a\sec^2 x + b\tan^2 x} = \frac{du}{a + (a + b)u^2}$ with $u = \tan x$.

With $s = \sqrt{\frac{a + b}{a}} > 0$: $F(x) = \frac{1}{as}\tan^{-1}(s\tan x)$, so $F\left(\frac{\pi}{4}\right) = \frac{\tan^{-1} s}{as}$ and $F\left(\frac{\pi}{3}\right) = \frac{\tan^{-1}(\sqrt{3}s)}{as}$.

Dividing the two conditions removes a: $\frac{\tan^{-1}(\sqrt{3}s)}{\tan^{-1} s} = \frac{\tan^{-1}(\sqrt{3}/2)}{\tan^{-1}(1/2)}$.

The left side decreases strictly from $\sqrt{3}$ (as $s \to 0^+$) towards $\frac{2}{3}$ (as $s \to \infty$), so the equation has exactly one root, and $s = \frac{1}{2}$ is that root.

Then $\frac{\tan^{-1}(1/2)}{a/2} = \frac{1}{2}\tan^{-1}\frac{1}{2}$ gives $a = 4$, and $\frac{a + b}{a} = s^2 = \frac{1}{4}$ gives $a + b = 1$, $b = -3$.

Check: $4 - 3\sin^2 x = \frac{5 + 3\cos 2x}{2}$, so $F(x) = G(2x)$ for all x, consistent with both conditions.

$a^2 + b^2 = 16 + 9 = 25$.

Final Answer: 25

Question 16

English

Let $F : \left(-\frac{\pi}{2}, \frac{5\pi}{6}\right) \to \mathbb{R}$ be a differentiable function such that $F(0) = 0$ and $F'(x) = \frac{1}{1 + \sin x + \sqrt{3}\cos x}$ for all $x \in \left(-\frac{\pi}{2}, \frac{5\pi}{6}\right)$. Let $\theta \in \left(-\frac{\pi}{2}, \frac{5\pi}{6}\right)$ be such that $F(\theta) = \frac{1}{\sqrt{3}}\log_e\frac{5 + 3\sqrt{3}}{2}$. Then the value of $\frac{180\,\theta}{\pi}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $F : \left(-\frac{\pi}{2}, \frac{5\pi}{6}\right) \to \mathbb{R}$ एक अवकलनीय फलन है जिसके लिए $F(0) = 0$ तथा सभी $x \in \left(-\frac{\pi}{2}, \frac{5\pi}{6}\right)$ के लिए $F'(x) = \frac{1}{1 + \sin x + \sqrt{3}\cos x}$ है। माना कि $\theta \in \left(-\frac{\pi}{2}, \frac{5\pi}{6}\right)$ ऐसा है कि $F(\theta) = \frac{1}{\sqrt{3}} \log_e \frac{5 + 3\sqrt{3}}{2}$ है। तब $\frac{180\,\theta}{\pi}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Put $t = \tan\frac{x}{2}$, so $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$, $dx = \frac{2\,dt}{1+t^2}$. For $x \in \left(-\frac{\pi}{2}, \frac{5\pi}{6}\right)$, $\frac{x}{2} \in \left(-\frac{\pi}{4}, \frac{5\pi}{12}\right)$ and $t \in (-1, 2+\sqrt{3})$.

$1 + \sin x + \sqrt{3}\cos x = \frac{(1+t^2) + 2t + \sqrt{3}(1-t^2)}{1+t^2} = \frac{(1-\sqrt{3})t^2 + 2t + (1+\sqrt{3})}{1+t^2}$.

The quadratic is the half-angle form of the equation $\sin x + \sqrt{3}\cos x = -1$. Its roots are $t = \frac{-2 \pm \sqrt{4 - 4(1-3)}}{2(1-\sqrt{3})} = \frac{-1 \pm \sqrt{3}}{1-\sqrt{3}}$, i.e. $t = -1$ and $t = 2+\sqrt{3}$; these are $x = -\frac{\pi}{2}$ and $x = \frac{5\pi}{6}$, the ends of the domain (equivalently $2\sin\left(x + \frac{\pi}{3}\right) = -1$).

So $(1-\sqrt{3})t^2 + 2t + (1+\sqrt{3}) = (1-\sqrt{3})(t+1)(t-2-\sqrt{3})$, and $F'(x)\,dx = \frac{2}{1-\sqrt{3}} \cdot \frac{dt}{(t+1)(t-2-\sqrt{3})}$.

Partial fractions: $\frac{1}{(t+1)(t-2-\sqrt{3})} = \frac{1}{3+\sqrt{3}}\left(\frac{1}{t-2-\sqrt{3}} - \frac{1}{t+1}\right)$, and $\frac{2}{(1-\sqrt{3})(3+\sqrt{3})} = \frac{2}{-2\sqrt{3}} = -\frac{1}{\sqrt{3}}$.

Hence $F = \frac{1}{\sqrt{3}} \log_e \left|\frac{t+1}{t-2-\sqrt{3}}\right| + C = \frac{1}{\sqrt{3}} \log_e \frac{t+1}{2+\sqrt{3}-t} + C$ on $-1 < t < 2+\sqrt{3}$.

$F(0) = 0$ $(t = 0)$ gives $C = \frac{1}{\sqrt{3}} \log_e (2+\sqrt{3})$, so $F(x) = \frac{1}{\sqrt{3}} \log_e \frac{(t+1)(2+\sqrt{3})}{2+\sqrt{3}-t}$.

The denominator $1 + 2\sin\left(x + \frac{\pi}{3}\right)$ is positive on the domain, so F is strictly increasing and θ is unique. The equation becomes $\frac{(t+1)(2+\sqrt{3})}{2+\sqrt{3}-t} = \frac{5+3\sqrt{3}}{2}$.

At $t = \sqrt{3}$: $\frac{(\sqrt{3}+1)(2+\sqrt{3})}{2} = \frac{2\sqrt{3}+3+2+\sqrt{3}}{2} = \frac{5+3\sqrt{3}}{2}$. So $\tan\frac{\theta}{2} = \sqrt{3}$, $\frac{\theta}{2} = \frac{\pi}{3}$ and $\theta = \frac{2\pi}{3}$.

Therefore $\frac{180\theta}{\pi} = 120$.

Final Answer: 120

Question 17

English

Let a and b be real numbers and let $f : (0, \infty) \to \mathbb{R}$ be defined by $f(x) = \dfrac{ax^2 + bx + 1}{x^2(x+1)^2}$. Suppose that f has an antiderivative $F : (0, \infty) \to \mathbb{R}$ that is a rational function, and that F has a local extremum at $x = 1$. Then the value of $a - b$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि a तथा b वास्तविक संख्याएँ हैं तथा $f : (0, \infty) \to \mathbb{R}$, $f(x) = \dfrac{ax^2 + bx + 1}{x^2(x+1)^2}$ द्वारा परिभाषित है। माना कि f का एक प्रतिअवकलज (antiderivative) $F : (0, \infty) \to \mathbb{R}$ एक परिमेय फलन है, तथा F का $x = 1$ पर एक स्थानीय चरम मान है। तब $a - b$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

f has double poles at 0 and -1. A rational antiderivative can only have poles there, of order one, so it has the form $F(x) = \dfrac{px + q}{x(x+1)} + c$; the numerator has degree less than 2 because $f = O\left(x^{-2}\right)$ at infinity.

Differentiate: $F'(x) = \dfrac{p\,x(x+1) - (px+q)(2x+1)}{x^2(x+1)^2} = \dfrac{px^2 + px - 2px^2 - px - 2qx - q}{x^2(x+1)^2} = \dfrac{-px^2 - 2qx - q}{x^2(x+1)^2}$.

Matching with $ax^2 + bx + 1$: constant term $-q = 1$, so $q = -1$; x-term $-2q = b$, so $b = 2$; x^2-term $-p = a$.

Cross-check by partial fractions: $f = \dfrac{b-2}{x} + \dfrac{1}{x^2} - \dfrac{b-2}{x+1} + \dfrac{a-b+1}{(x+1)^2}$. The terms $\dfrac{b-2}{x}$ and $\dfrac{b-2}{x+1}$ would produce $\log_e$ terms, so a rational antiderivative requires $b = 2$, the same condition.

$F'(1) = f(1) = 0$ is necessary for a local extremum: $\dfrac{a+2+1}{4} = 0$, so $a = -3$.

Then $f(x) = \dfrac{-3x^2 + 2x + 1}{x^2(x+1)^2} = \dfrac{(1-x)(1+3x)}{x^2(x+1)^2}$, positive on $(0, 1)$ and negative on $(1, \infty)$, so F has a local maximum at $x = 1$ and the condition holds. Here $F(x) = -\dfrac{1}{x} + \dfrac{4}{x+1} + c$.

Hence $a - b = -3 - 2 = -5$.

Final Answer: -5

Question 18

English

Let I be an open interval containing 1 and let $f : I \to (0, 1)$ be a differentiable function with $f(1) = \dfrac{1}{3}$ such that $y = f(x)$ satisfies the differential equation

$$\left(y + y^3 - x^2y^2\right) dx + \left(x - xy^2\right) dy = 0$$

for all $x \in I$. If $\dfrac{3}{2} \in I$, then the value of $f\left(\dfrac{3}{2}\right)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि I एक विवृत अन्तराल है जिसमें 1 है, तथा $f : I \to (0,1)$ एक अवकलनीय फलन है जिसके लिए $f(1) = \dfrac{1}{3}$ है और $y = f(x)$ सभी $x \in I$ के लिए अवकल समीकरण

$$\left(y + y^3 - x^2y^2\right) dx + \left(x - xy^2\right) dy = 0$$

को सन्तुष्ट करता है। यदि $\dfrac{3}{2} \in I$ है, तो $f\left(\dfrac{3}{2}\right)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The equation is not separable and not exact as written. Expand and regroup: $y\,dx + x\,dy + y^3\,dx - xy^2\,dy - x^2y^2\,dx = (y\,dx + x\,dy) + y^2(y\,dx - x\,dy) - x^2y^2\,dx$.

Recognise the perfect differentials $y\,dx + x\,dy = d(xy)$ and $x\,dy - y\,dx = x^2\,d\left(\dfrac{y}{x}\right)$, so $y^2(y\,dx - x\,dy) = -x^2y^2\,d\left(\dfrac{y}{x}\right)$.

The equation becomes $d(xy) = x^2y^2\left[d\left(\dfrac{y}{x}\right) + dx\right]$. Since $x \neq 0$ and $y \neq 0$ near $(1, \frac{1}{3})$, divide by $(xy)^2$ (integrating factor $\dfrac{1}{x^2y^2}$ found by inspection): $\dfrac{d(xy)}{(xy)^2} = d\left(\dfrac{y}{x}\right) + dx$.

Each side is exact: $-\dfrac{1}{xy} = \dfrac{y}{x} + x + C$.

At $x = 1$, $y = \dfrac{1}{3}$: $-3 = \dfrac{1}{3} + 1 + C$, so $C = -\dfrac{13}{3}$. Multiplying by $-xy$: $1 = -y^2 - x^2y + \dfrac{13}{3}xy$, i.e. $y + \dfrac{1}{y} = \dfrac{13x}{3} - x^2$.

Let $g(x) = \dfrac{13x}{3} - x^2$. On $\left[1, \dfrac{3}{2}\right]$, g increases from $\dfrac{10}{3}$ to $\dfrac{17}{4}$, so $g > 2$ and the branch $y = \dfrac{g - \sqrt{g^2 - 4}}{2} \in (0,1)$ is a differentiable solution there; the coefficient $x - xy^2$ stays non-zero because $y < 1$.

At $x = \dfrac{3}{2}$: $y + \dfrac{1}{y} = \dfrac{17}{4}$, so $4y^2 - 17y + 4 = 0$, $(4y-1)(y-4) = 0$. Since $0 < f < 1$, $f\left(\dfrac{3}{2}\right) = \dfrac{1}{4} = 0.25$.

Final Answer: 0.25